IT2010 – Mobile Application Development
BSc (Hons) in Information Technology
2nd Year
Faculty of Computing
SLIIT

2025 – Lab Exam 03
Report

Student ID	IT23224384
Batch	Y2.S2.WE.IT.03.02
Marking Guide	
1. **Functionality**: How well the core and bonus features are implemented	**4**
2. **Creativity & Usability**: Clean and intuitive UI/UX design	**2**
3. **Validation & Error Handling**: Proper input validation, error handling, and user feedback implementation	**2**
4. **Data Persistence**: Proper use of **SharedPreferences** and **Internal Storage**	**2**
Total Marks	**10**
Evaluator	

Description:

Application Name

FINORA – TRACK YOUR EXPENSES

Overview

Finora is an innovative and user-friendly personal finance management application tailored for individuals who wish to gain control over their finances without the intricacies associated with conventional budgeting methods. Suitable for students, professionals, entrepreneurs, and homemakers alike, Finora enables users to oversee their income, monitor expenditures, establish savings objectives, and acquire insights into their financial behaviors—accessible at any time and from any location. The application ensures a smooth user experience by providing personalized budgeting strategies, real-time expense monitoring, financial goal establishment, and intelligent insights tailored to individual preferences. Finora removes the necessity for

spreadsheets or complicated tools, allowing users to enhance their savings, curb excessive spending, and maintain financial oversight effortlessly. With functionalities such as visual representations of expenses, customized budget notifications, and monthly financial summaries, users can cultivate better financial practices conveniently through their mobile devices. Finora delivers financial transparency and autonomy directly to your hands.

First Time Using the App

New users can sign up and log in to unlock the full potential of **Finora.** Upon logging in, users will be prompted to enter their income sources, recurring expenses, and savings goals to generate a tailored financial plan.

Using the App

After account creation and login, users are directed to the **dashboard**, where they can explore spending trends, track budgets, and monitor their progress. Users can log expenses, categorize transactions, and receive spending alerts or reminders.

Finora **enables users to:**

✓ Track daily, weekly, and monthly expenses with ease
✓ Categorize income and spending (e.g., food, bills, transport)
✓ Set personalized savings goals and monitor progress
✓ View reports and charts for better financial decision-making
✓ Stay motivated through savings milestones and challenges

Finora offers a clear, flexible path to financial wellness, allowing users to manage their money and meet their goals without stress.

Problem Statement

In today's fast-paced world, managing personal finances is often overlooked. Many individuals struggle with tracking expenses, overspending, and achieving savings goals due to the lack of time, knowledge, or access to effective tools. Traditional methods like notebooks or spreadsheets are inefficient and tedious, while many finance apps lack simplicity or personalization.

Finora addresses this gap by providing a modern, user-friendly solution for everyday money management. By offering personalized budgets, visual analytics, flexible tracking, and smart reminders, Finora makes financial health easy, engaging, and achievable. It empowers users to build better financial habits, gain clarity, and move toward their money goals confidently.

Target Audience

Finora is tailored to support a wide range of users who seek financial clarity and simplicity:

✓ **Busy Professionals** – Individuals managing tight schedules who need a quick, efficient way to handle budgets and expenses.
✓ **Students** – Young adults aiming to track allowances, part-time income, and spending without complex tools.
✓ **Stay-at-Home Parents** – Parents who want to manage household budgets and stay on top of family expenses.
✓ **Freelancers & Entrepreneurs** – Independent earners looking to categorize income, monitor business spending, and plan savings.
✓ **Budgeting Beginners** – Anyone new to money management needing a friendly, guided finance tracker.

Key Features

• Smart Budgeting (Main Function)

Create customized budgets based on income, spending habits, and savings goals.
Personalized spending insights to keep you on track.
Easy-to-understand visual breakdowns of your finances.

• Expense & Income Tracker

Quickly log transactions and categorize them (food, bills, travel, etc.).
Add notes or tags for detailed tracking.
Monitor recurring expenses and alerts for bill payments.

• Goal-Oriented Saving Plans

Set and track financial goals like emergency funds, vacation savings, or debt repayment.
See real-time progress and adjust contributions easily.
Encouraging milestones to help you stay motivated.

• Financial Insights & Recommendations

Smart insights based on your habits and budget behavior.
Suggested adjustments to reduce overspending or improve savings.
Weekly reports for better decision-making.

• Reminders & Alerts

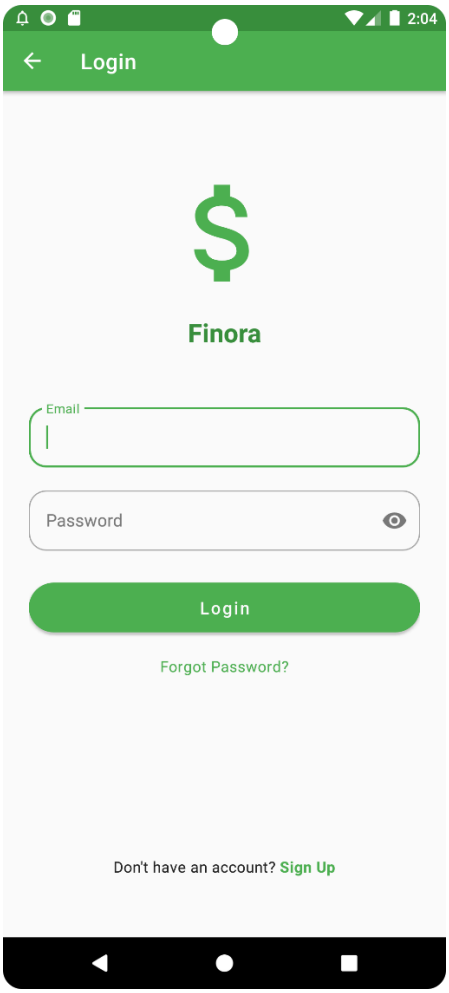
Bill due reminders, low-balance alerts, and custom spending limit warnings.
Daily and monthly summaries to keep you updated.

• **Data Sync & Security**

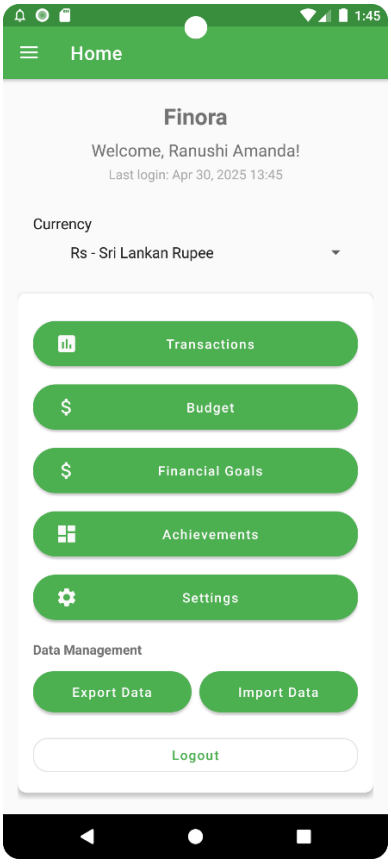
Sync across devices and cloud backup.
Your financial data stays private with bank-grade encryption and secure local storage.

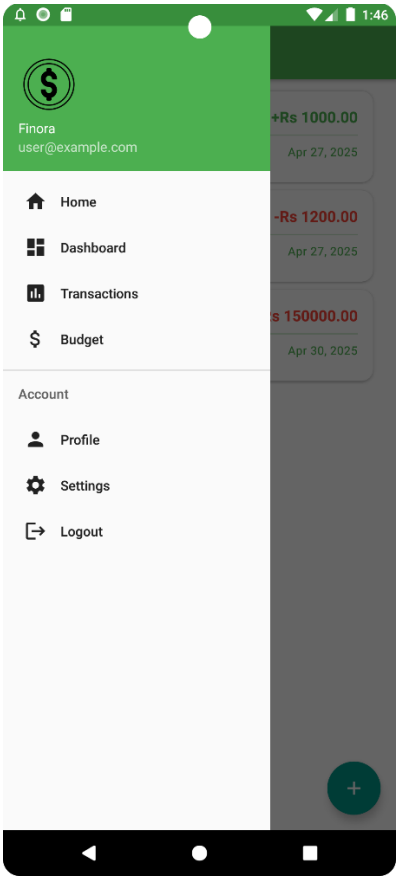
Screenshots:

Login Page:

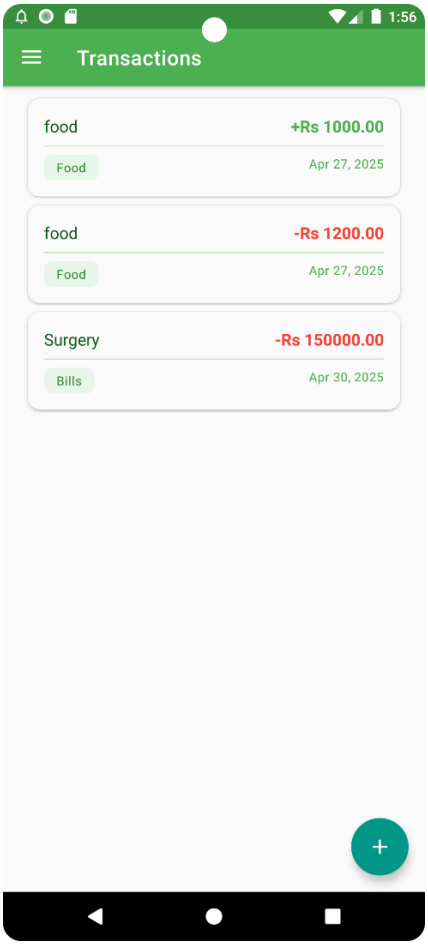


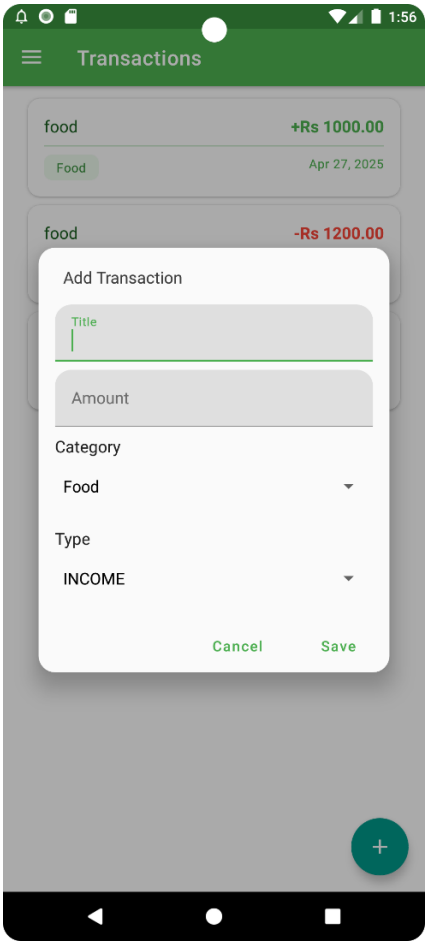
Home Screen:


1:45
Home
Finora
Welcome, Ranushi Amanda!
Last login: Apr 30, 2025 13:45
Currency
Rs - Sri Lankan Rupee
Transactions
Budget
Financial Goals
Achievements
Settings
Data Management
Export Data
Import Data
Logout

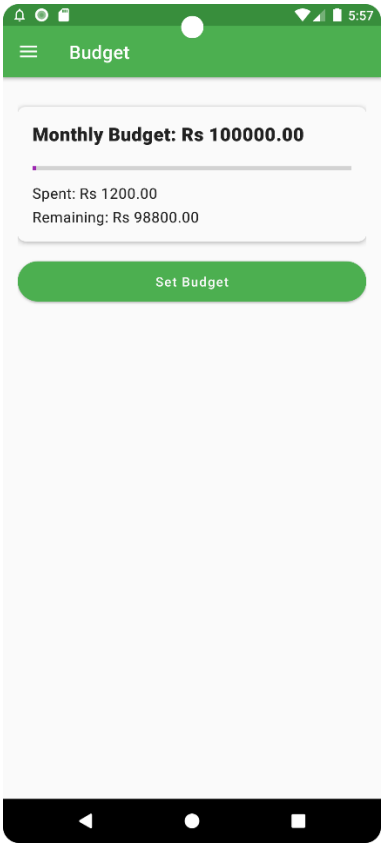

1:46
Finora
user@example.com
Home
Dashboard
Transactions
Budget
Account
Profile
Settings
Logout
+Rs 1000.00
Apr 27, 2025
-Rs 1200.00
Apr 27, 2025
Apr 30, 2025

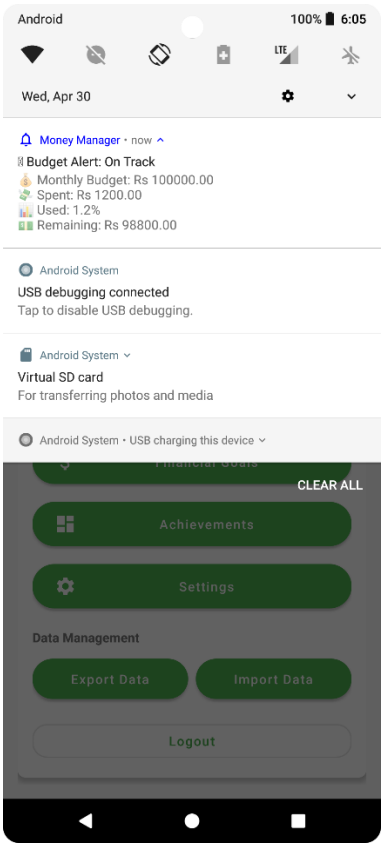
Transactions:



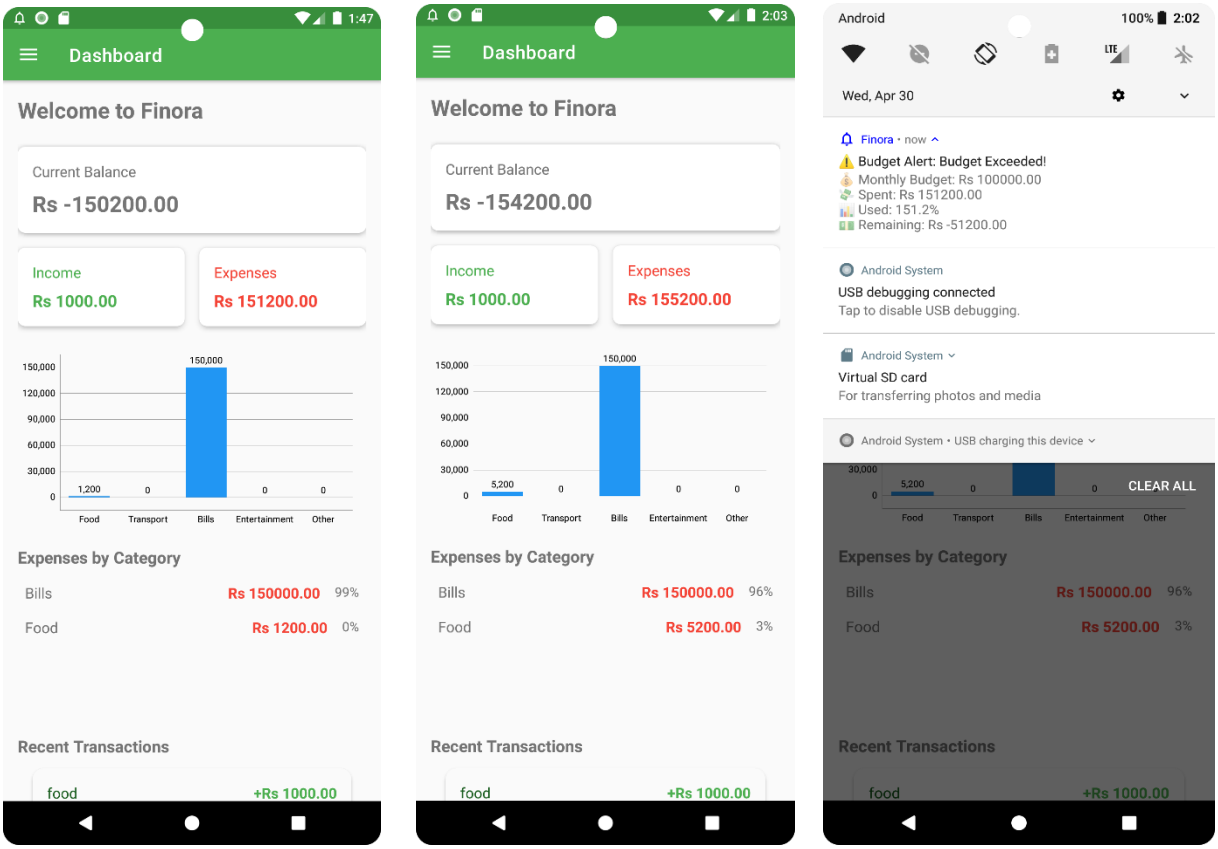


Budget Set:

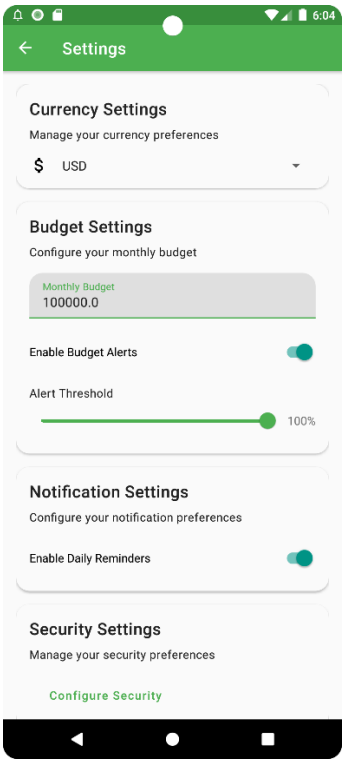

5:57
Budget
Monthly Budget: Rs 100000.00
Spent: Rs 1200.00
Remaining: Rs 98800.00
Set Budget

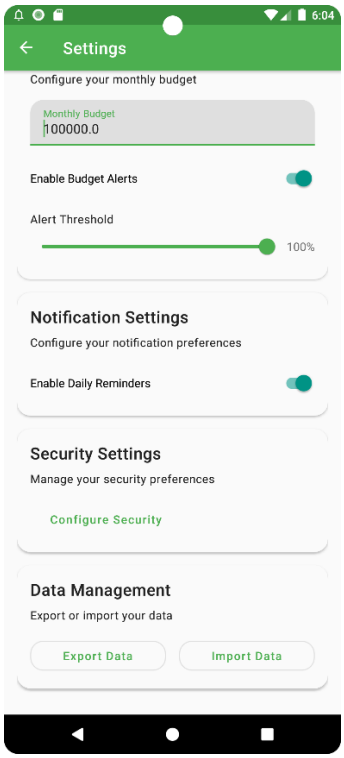

Android
100% 6:05
Wed, Apr 30
Money Manager · now
Budget Alert: On Track
Monthly Budget: Rs 100000.00
Spent: Rs 1200.00
Used: 1.2%
Remaining: Rs 98800.00
Android System
USB debugging connected
Tap to disable USB debugging.
Android System
Virtual SD card
For transferring photos and media
Android System · USB charging this device
CLEAR ALL
Achievements
Settings
Data Management
Export Data
Import Data
Logout

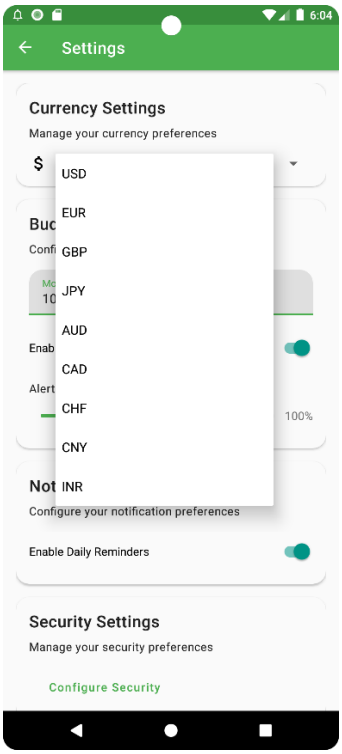
Dashboard:

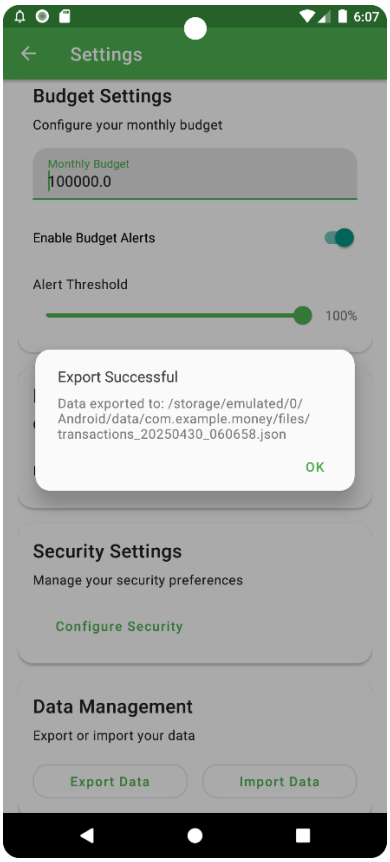

1:47
Dashboard
Welcome to Finora
Current Balance
Rs -150200.00
Income
Rs 1000.00
Expenses
Rs 151200.00
150,000
120,000
90,000
60,000
30,000
0
1,200
0
150,000
0
0
Food
Transport
Bills
Entertainment
Other
Expenses by Category
Bills
Rs 150000.00
99%
Food
Rs 1200.00
0%
Recent Transactions
food
+Rs 1000.00
2:03
Dashboard
Welcome to Finora
Current Balance
Rs -154200.00
Income
Rs 1000.00
Expenses
Rs 155200.00
150,000
120,000
90,000
60,000
30,000
0
5,200
0
150,000
0
0
Food
Transport
Bills
Entertainment
Other
Expenses by Category
Bills
Rs 150000.00
96%
Food
Rs 5200.00
3%
Recent Transactions
food
+Rs 1000.00
Android
100% 2:02
Wed, Apr 30
Finora • now
Budget Alert: Budget Exceeded!
Monthly Budget: Rs 100000.00
Spent: Rs 151200.00
Used: 151.2%
Remaining: Rs -51200.00
Android System
USB debugging connected
Tap to disable USB debugging.
Android System
Virtual SD card
For transferring photos and media
Android System • USB charging this device
30,000
0
5,200
0
0
CLEAR ALL
Food
Transport
Bills
Entertainment
Other
Expenses by Category
Bills
Rs 150000.00
96%
Food
Rs 5200.00
3%
Recent Transactions
food
+Rs 1000.00

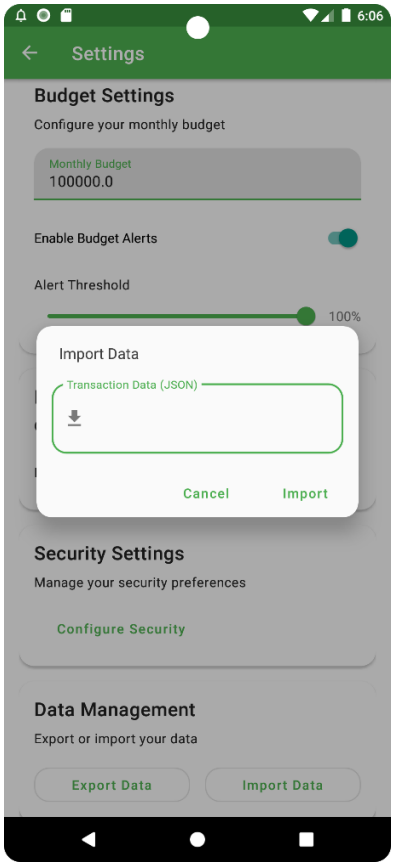
Settings:

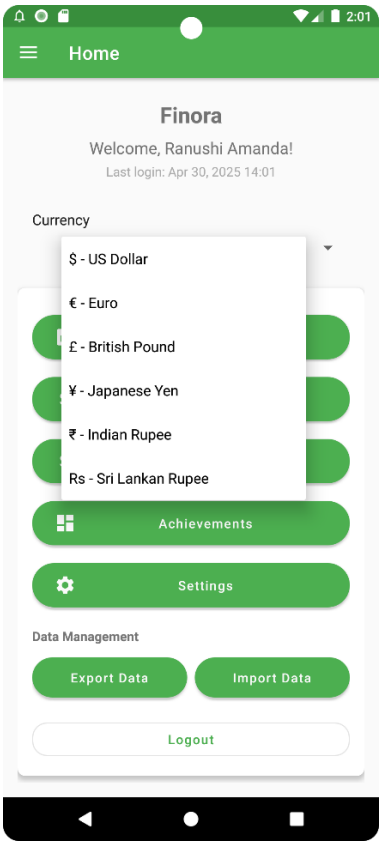

6:04
Settings
Currency Settings
Manage your currency preferences
$ USD
Budget Settings
Configure your monthly budget
Monthly Budget
100000.0
Enable Budget Alerts
Alert Threshold
100%
Notification Settings
Configure your notification preferences
Enable Daily Reminders
Security Settings
Manage your security preferences
Configure Security

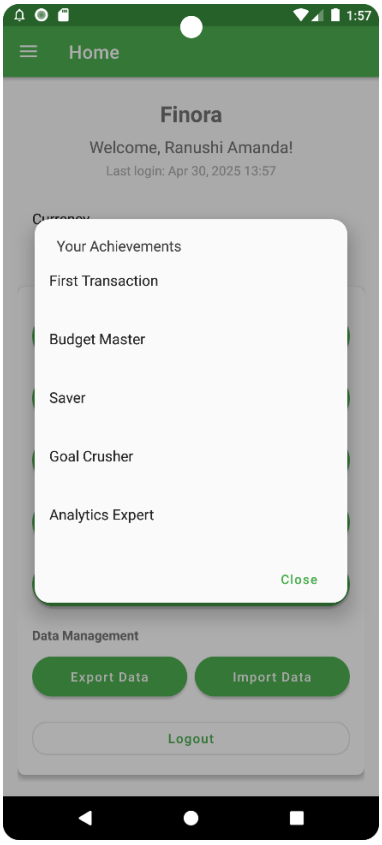

6:04
Settings
Configure your monthly budget
Monthly Budget
100000.0
Enable Budget Alerts
Alert Threshold
100%
Notification Settings
Configure your notification preferences
Enable Daily Reminders
Security Settings
Manage your security preferences
Configure Security
Data Management
Export or import your data
Export Data
Import Data

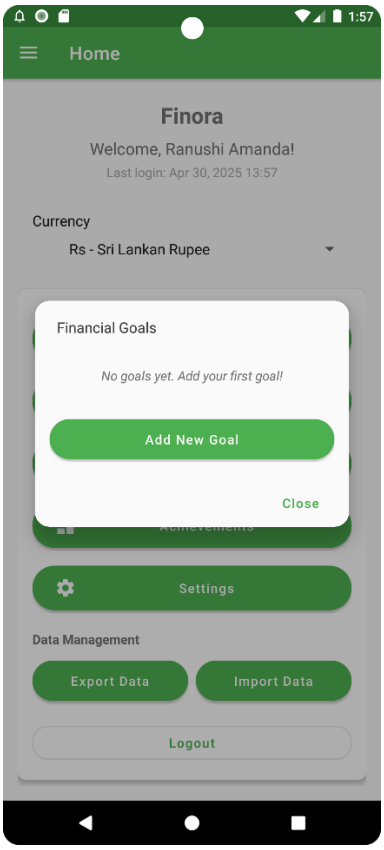

6:04
Settings
Currency Settings
Manage your currency preferences
$
USD
EUR
GBP
JPY
AUD
CAD
CHF
CNY
INR
Configure your notification preferences
Enable Daily Reminders
Security Settings
Manage your security preferences
Configure Security

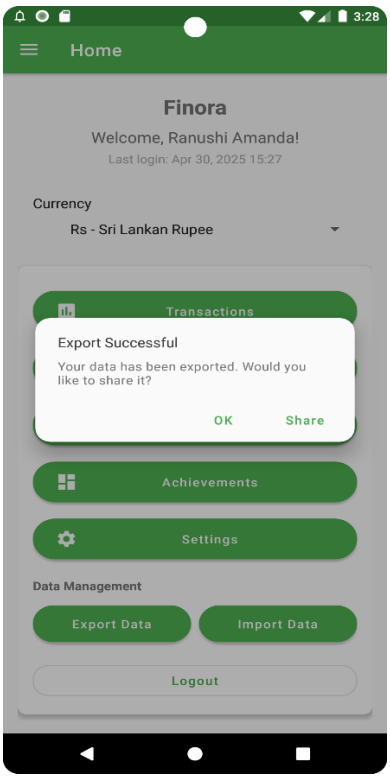

6:07
Settings
Budget Settings
Configure your monthly budget
Monthly Budget
100000.0
Enable Budget Alerts
Alert Threshold
100%
Export Successful
Data exported to: /storage/emulated/0/
Android/data/com.example.money/files/
transactions_20250430_060658.json
OK
Security Settings
Manage your security preferences
Configure Security
Data Management
Export or import your data
Export Data
Import Data


6:06
Settings
Budget Settings
Configure your monthly budget
Monthly Budget
100000.0
Enable Budget Alerts
Alert Threshold
100%
Import Data
Transaction Data (JSON)
Cancel
Import
Security Settings
Manage your security preferences
Configure Security
Data Management
Export or import your data
Export Data
Import Data

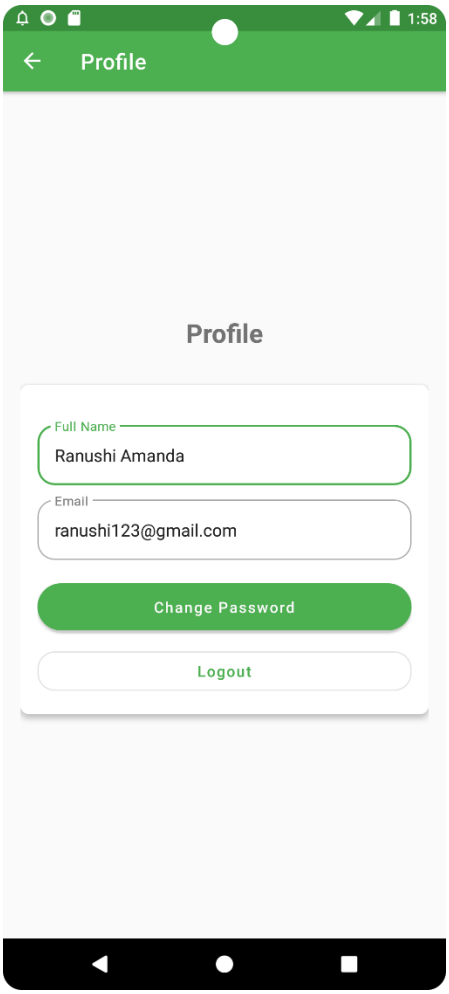

2:01
Home
Finora
Welcome, Ranushi Amanda!
Last login: Apr 30, 2025 14:01
Currency
$ - US Dollar
€ - Euro
£ - British Pound
¥ - Japanese Yen
₹ - Indian Rupee
Rs - Sri Lankan Rupee
Achievements
Settings
Data Management
Export Data
Import Data
Logout


1:57
Home
Finora
Welcome, Ranushi Amanda!
Last login: Apr 30, 2025 13:57
Your Achievements
First Transaction
Budget Master
Saver
Goal Crusher
Analytics Expert
Close
Data Management
Export Data
Import Data
Logout


1:57
Home
Finora
Welcome, Ranushi Amanda!
Last login: Apr 30, 2025 13:57
Currency
Rs - Sri Lankan Rupee
Financial Goals
No goals yet. Add your first goal!
Add New Goal
Close
Settings
Data Management
Export Data
Import Data
Logout

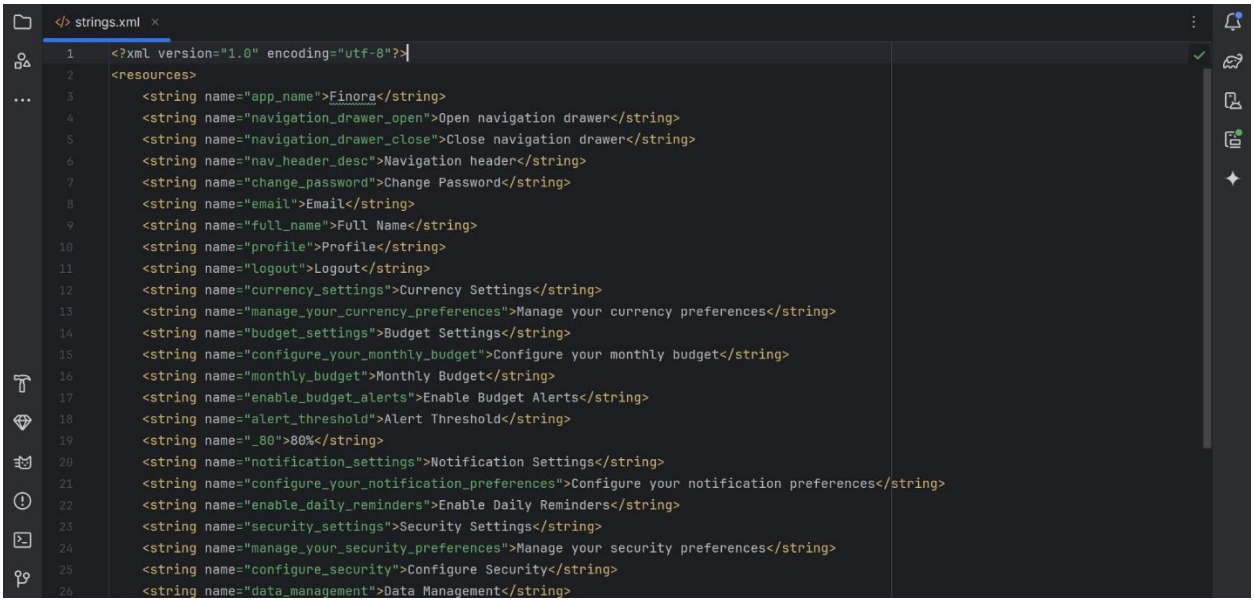

3:28
Home
Finora
Welcome, Ranushi Amanda!
Last login: Apr 30, 2025 15:27
Currency
Rs - Sri Lankan Rupee
Transactions
Export Successful
Your data has been exported. Would you like to share it?
OK
Share
Achievements
Settings
Data Management
Export Data
Import Data
Logout

Profile:

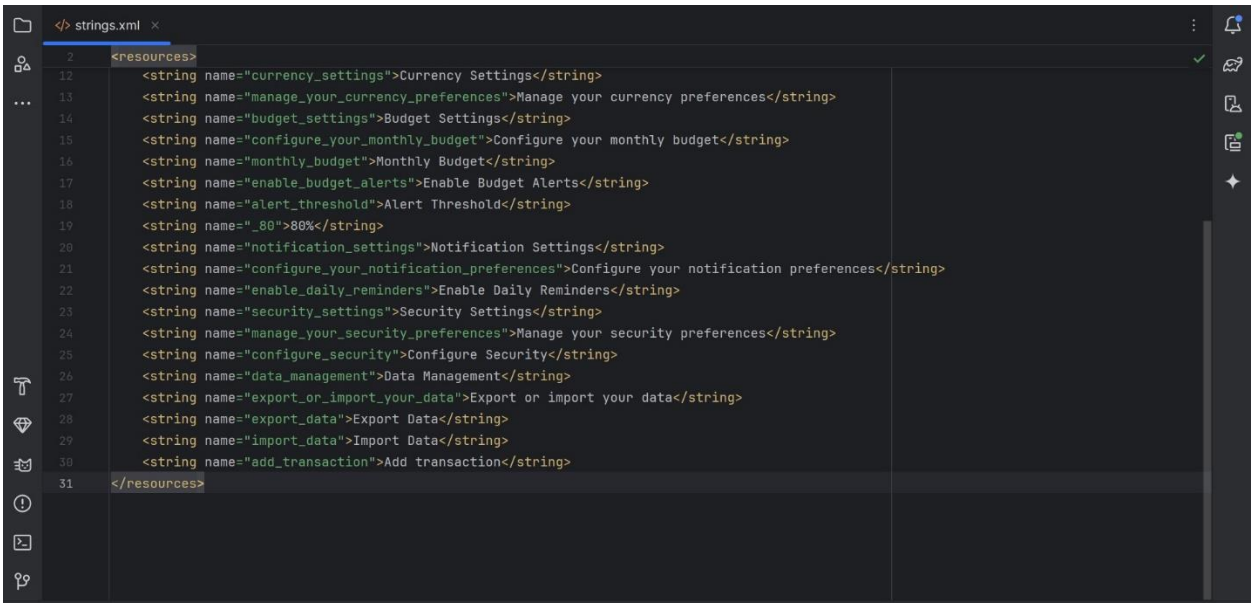

1:58
Profile
Profile
Full Name
Ranushi Amanda
Email
ranushi123@gmail.com
Change Password
Logout

Content of xml files of Strings and Colors:

Strings



```xml
<?xml version="1.0" encoding="utf-8"?>
<resources>
    <string name="app_name">Finora</string>
    <string name="navigation_drawer_open">Open navigation drawer</string>
    <string name="navigation_drawer_close">Close navigation drawer</string>
    <string name="nav_header_desc">Navigation header</string>
    <string name="change_password">Change Password</string>
    <string name="email">Email</string>
    <string name="full_name">Full Name</string>
    <string name="profile">Profile</string>
    <string name="logout">Logout</string>
    <string name="currency_settings">Currency Settings</string>
    <string name="manage_your_currency_preferences">Manage your currency preferences</string>
    <string name="budget_settings">Budget Settings</string>
    <string name="configure_your_monthly_budget">Configure your monthly budget</string>
    <string name="monthly_budget">Monthly Budget</string>
    <string name="enable_budget_alerts">Enable Budget Alerts</string>
    <string name="alert_threshold">Alert Threshold</string>
    <string name="_80">80%</string>
    <string name="notification_settings">Notification Settings</string>
    <string name="configure_your_notification_preferences">Configure your notification preferences</string>
    <string name="enable_daily_reminders">Enable Daily Reminders</string>
    <string name="security_settings">Security Settings</string>
    <string name="manage_your_security_preferences">Manage your security preferences</string>
    <string name="configure_security">Configure Security</string>
    <string name="data_management">Data Management</string>
```



```xml
<resources>
    <string name="currency_settings">Currency Settings</string>
    <string name="manage_your_currency_preferences">Manage your currency preferences</string>
    <string name="budget_settings">Budget Settings</string>
    <string name="configure_your_monthly_budget">Configure your monthly budget</string>
    <string name="monthly_budget">Monthly Budget</string>
    <string name="enable_budget_alerts">Enable Budget Alerts</string>
    <string name="alert_threshold">Alert Threshold</string>
    <string name="_80">80%</string>
    <string name="notification_settings">Notification Settings</string>
    <string name="configure_your_notification_preferences">Configure your notification preferences</string>
    <string name="enable_daily_reminders">Enable Daily Reminders</string>
    <string name="security_settings">Security Settings</string>
    <string name="manage_your_security_preferences">Manage your security preferences</string>
    <string name="configure_security">Configure Security</string>
    <string name="data_management">Data Management</string>
    <string name="export_or_import_your_data">Export or import your data</string>
    <string name="export_data">Export Data</string>
    <string name="import_data">Import Data</string>
    <string name="add_transaction">Add transaction</string>
</resources>
```

Colors

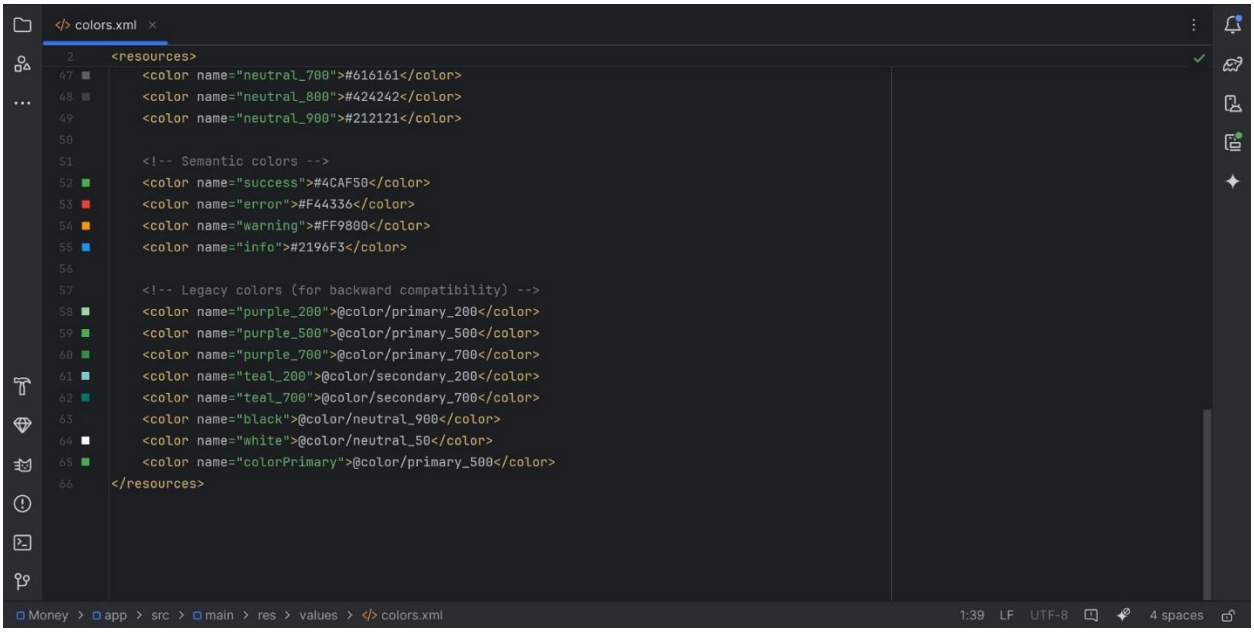
colors.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<resources>
    <!-- Primary colors - -->
    <color name="primary_50">#E8F5E9</color>
    <color name="primary_100">#C8E6C9</color>
    <color name="primary_200">#A5D6A7</color>
    <color name="primary_300">#81C784</color>
    <color name="primary_400">#66BB6A</color>
    <color name="primary_500">#4CAF50</color>
    <color name="primary_600">#43A047</color>
    <color name="primary_700">#388E3C</color>
    <color name="primary_800">#2E7D32</color>
    <color name="primary_900">#1B5E20</color>

    <!-- Secondary colors  -->
    <color name="secondary_50">#E0F2F1</color>
    <color name="secondary_100">#B2DFDB</color>
    <color name="secondary_200">#80CBC4</color>
    <color name="secondary_300">#4DB6AC</color>
    <color name="secondary_400">#26A69A</color>
    <color name="secondary_500">#009688</color>
    <color name="secondary_600">#00897B</color>
    <color name="secondary_700">#00796B</color>
    <color name="secondary_800">#00695C</color>
    <color name="secondary_900">#004D40</color>
```

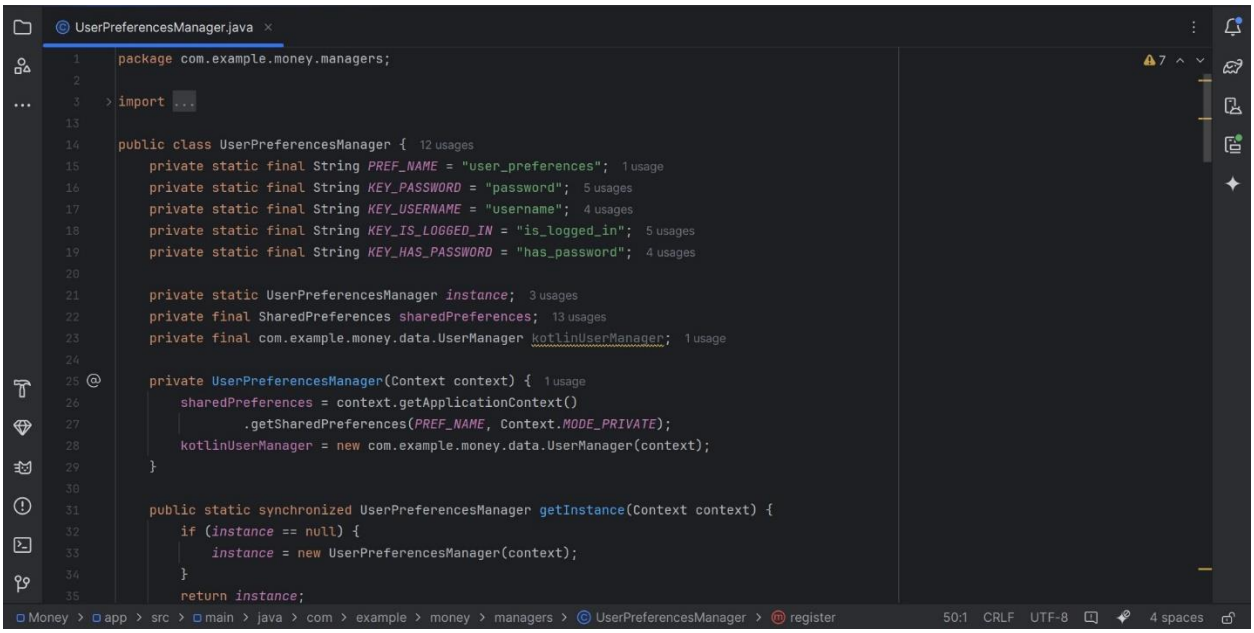
colors.xml

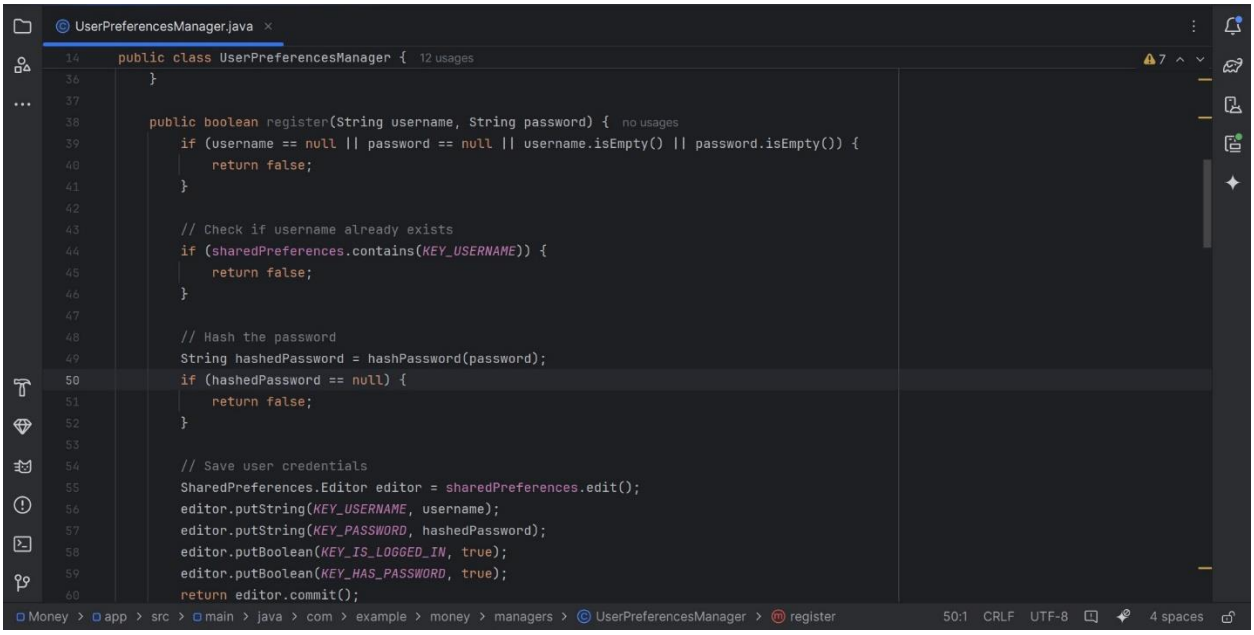
```xml
<resources>
    <!-- Accent colors - Coral -->
    <color name="accent_50">#FFEBEE</color>
    <color name="accent_100">#FFCDD2</color>
    <color name="accent_200">#EF9A9A</color>
    <color name="accent_300">#E57373</color>
    <color name="accent_400">#EF5350</color>
    <color name="accent_500">#F44336</color>
    <color name="accent_600">#E53935</color>
    <color name="accent_700">#D32F2F</color>
    <color name="accent_800">#C62828</color>
    <color name="accent_900">#B71C1C</color>

    <!-- Neutral colors - Modern Gray -->
    <color name="neutral_50">#FAFAFA</color>
    <color name="neutral_100">#F5F5F5</color>
    <color name="neutral_200">#EEEEEE</color>
    <color name="neutral_300">#E0E0E0</color>
    <color name="neutral_400">#BDBDBD</color>
    <color name="neutral_500">#9E9E9E</color>
    <color name="neutral_600">#757575</color>
    <color name="neutral_700">#616161</color>
    <color name="neutral_800">#424242</color>
    <color name="neutral_900">#212121</color>

    <!-- Semantic colors -->
```

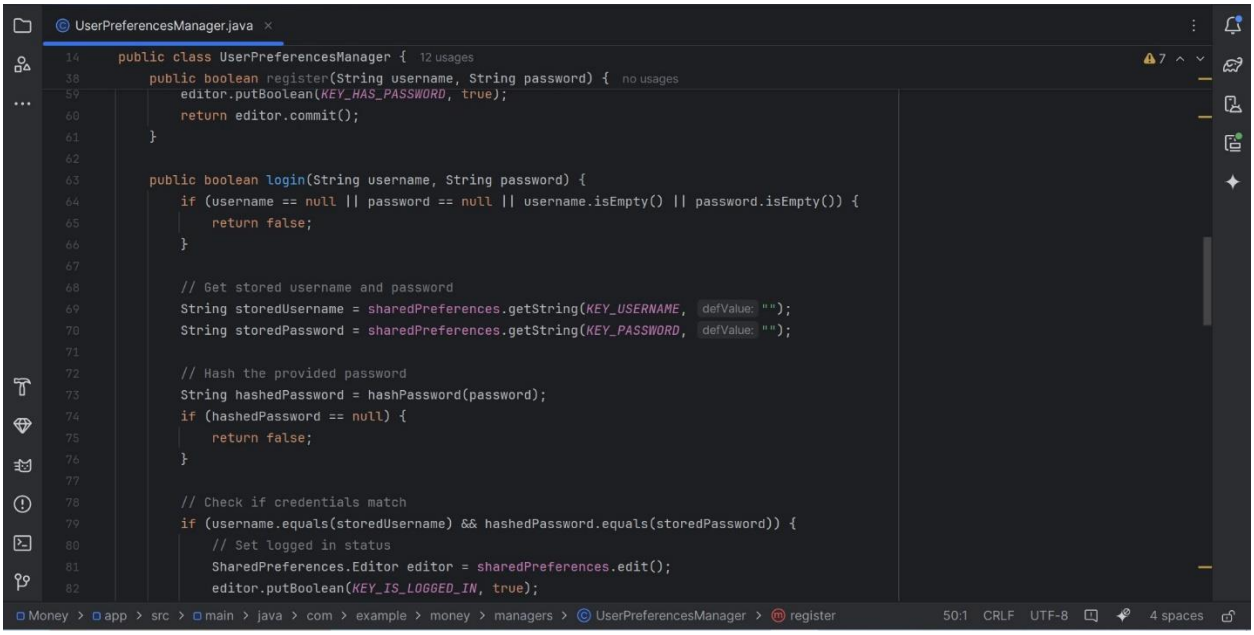
Money › app › src › main › res › values › colors.xml 1:39 LF UTF-8 4 spaces

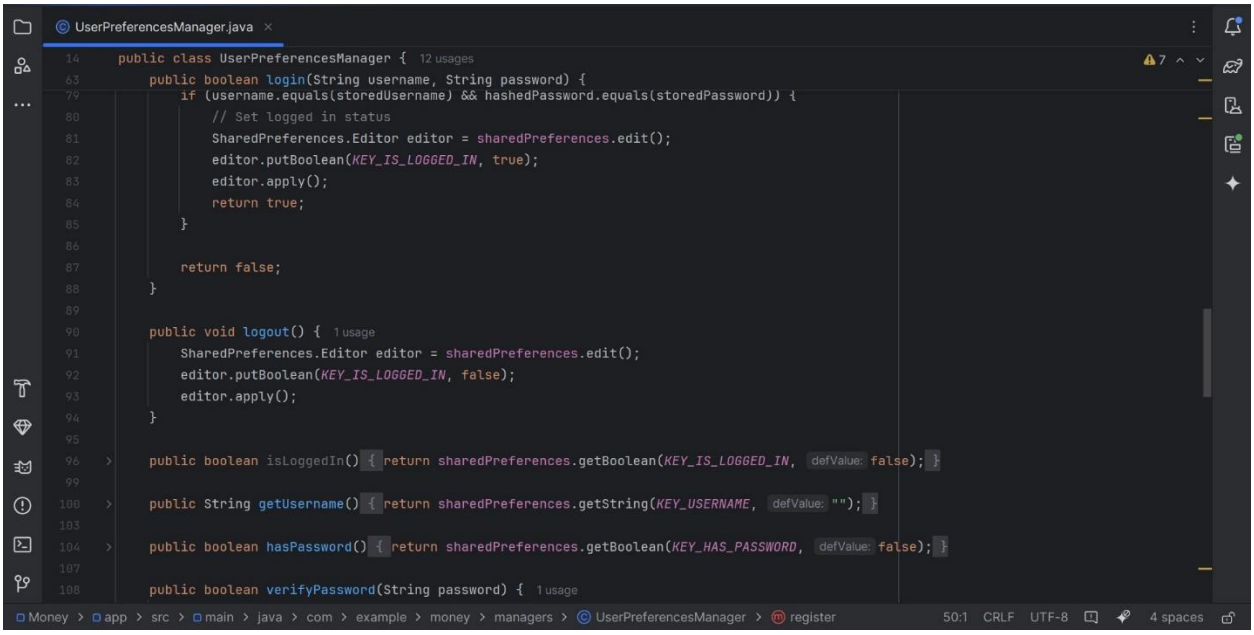


User Preferences:




UserPreferencesManager.java
public class UserPreferencesManager { 12 usages
}
public boolean register(String username, String password) { no usages
if (username == null || password == null || username.isEmpty() || password.isEmpty()) {
return false;
}
// Check if username already exists
if (sharedPreferences.contains(KEY_USERNAME)) {
return false;
}
// Hash the password
String hashedPassword = hashPassword(password);
if (hashedPassword == null) {
return false;
}
// Save user credentials
SharedPreferences.Editor editor = sharedPreferences.edit();
editor.putString(KEY_USERNAME, username);
editor.putString(KEY_PASSWORD, hashedPassword);
editor.putBoolean(KEY_IS_LOGGED_IN, true);
editor.putBoolean(KEY_HAS_PASSWORD, true);
return editor.commit();
Money > app > src > main > java > com > example > money > managers > UserPreferencesManager > register
50:1 CRLF UTF-8 4 spaces


UserPreferencesManager.java
public class UserPreferencesManager { 12 usages
public boolean register(String username, String password) { no usages
editor.putBoolean(KEY_HAS_PASSWORD, true);
return editor.commit();
}
public boolean login(String username, String password) {
if (username == null || password == null || username.isEmpty() || password.isEmpty()) {
return false;
}
// Get stored username and password
String storedUsername = sharedPreferences.getString(KEY_USERNAME, defValue: "");
String storedPassword = sharedPreferences.getString(KEY_PASSWORD, defValue: "");
// Hash the provided password
String hashedPassword = hashPassword(password);
if (hashedPassword == null) {
return false;
}
// Check if credentials match
if (username.equals(storedUsername) && hashedPassword.equals(storedPassword)) {
// Set logged in status
SharedPreferences.Editor editor = sharedPreferences.edit();
editor.putBoolean(KEY_IS_LOGGED_IN, true);
Money > app > src > main > java > com > example > money > managers > UserPreferencesManager > register
50:1 CRLF UTF-8 4 spaces


UserPreferencesManager.java
public class UserPreferencesManager { 12 usages
public boolean login(String username, String password) {
if (username.equals(storedUsername) && hashedPassword.equals(storedPassword)) {
// Set logged in status
SharedPreferences.Editor editor = sharedPreferences.edit();
editor.putBoolean(KEY_IS_LOGGED_IN, true);
editor.apply();
return true;
}
return false;
}
public void logout() { 1 usage
SharedPreferences.Editor editor = sharedPreferences.edit();
editor.putBoolean(KEY_IS_LOGGED_IN, false);
editor.apply();
}
public boolean isLoggedIn() { return sharedPreferences.getBoolean(KEY_IS_LOGGED_IN, defValue: false); }
public String getUsername() { return sharedPreferences.getString(KEY_USERNAME, defValue: ""); }
public boolean hasPassword() { return sharedPreferences.getBoolean(KEY_HAS_PASSWORD, defValue: false); }
public boolean verifyPassword(String password) { 1 usage
Money > app > src > main > java > com > example > money > managers > UserPreferencesManager > register
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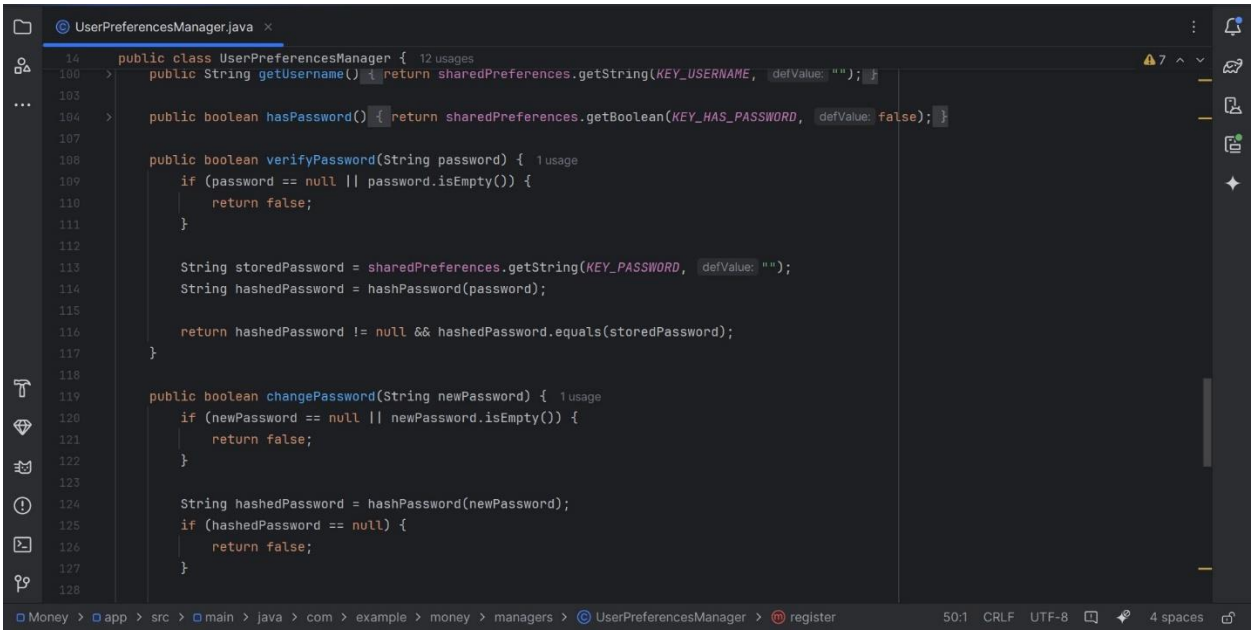
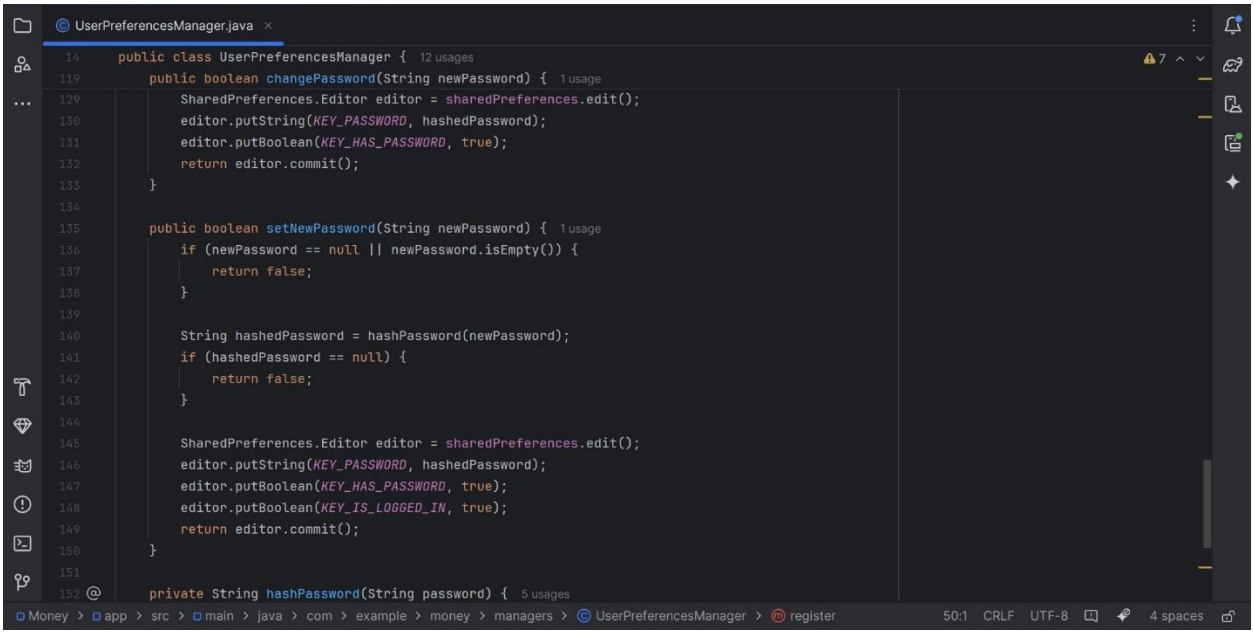
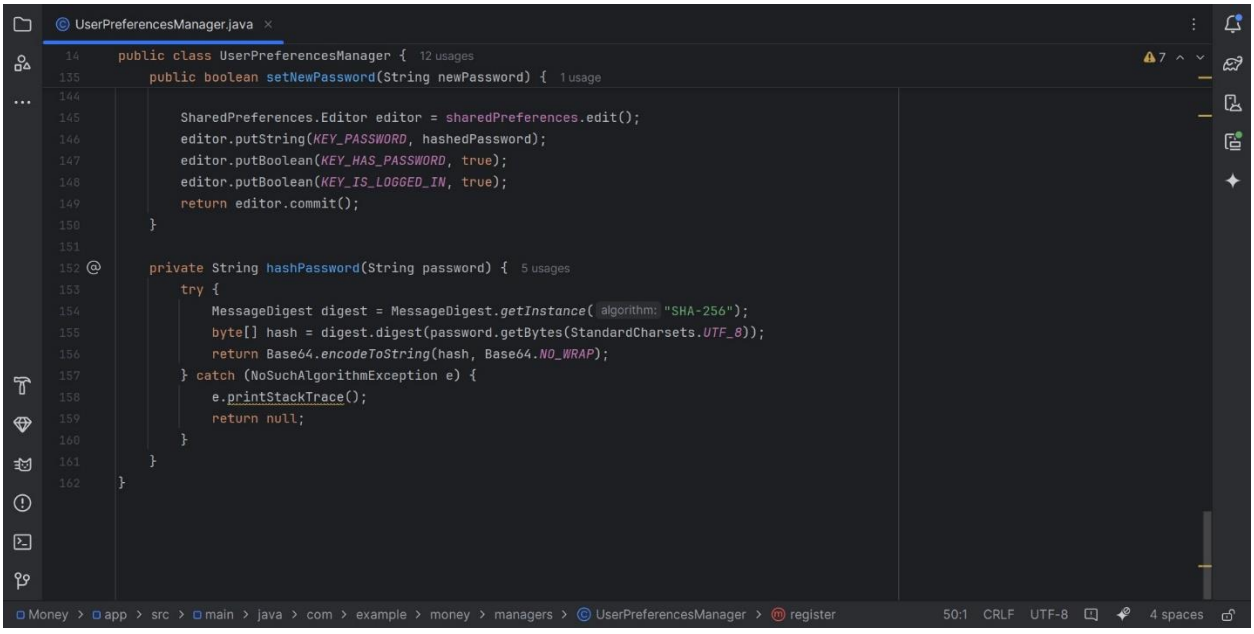

UserPreferencesManager.java
public class UserPreferencesManager { 12 usages
public String getUsername() { return sharedPreferences.getString(KEY_USERNAME, defValue: ""); }
public boolean hasPassword() { return sharedPreferences.getBoolean(KEY_HAS_PASSWORD, defValue: false); }
public boolean verifyPassword(String password) { 1 usage
if (password == null || password.isEmpty()) {
return false;
}
String storedPassword = sharedPreferences.getString(KEY_PASSWORD, defValue: "");
String hashedPassword = hashPassword(password);
return hashedPassword != null && hashedPassword.equals(storedPassword);
}
public boolean changePassword(String newPassword) { 1 usage
if (newPassword == null || newPassword.isEmpty()) {
return false;
}
String hashedPassword = hashPassword(newPassword);
if (hashedPassword == null) {
return false;
}
Money > app > src > main > java > com > example > money > managers > UserPreferencesManager > register
50:1 CRLF UTF-8 4 spaces


UserPreferencesManager.java
public class UserPreferencesManager { 12 usages
public boolean changePassword(String newPassword) { 1 usage
SharedPreferences.Editor editor = sharedPreferences.edit();
editor.putString(KEY_PASSWORD, hashedPassword);
editor.putBoolean(KEY_HAS_PASSWORD, true);
return editor.commit();
}
public boolean setNewPassword(String newPassword) { 1 usage
if (newPassword == null || newPassword.isEmpty()) {
return false;
}
String hashedPassword = hashPassword(newPassword);
if (hashedPassword == null) {
return false;
}
SharedPreferences.Editor editor = sharedPreferences.edit();
editor.putString(KEY_PASSWORD, hashedPassword);
editor.putBoolean(KEY_HAS_PASSWORD, true);
editor.putBoolean(KEY_IS_LOGGED_IN, true);
return editor.commit();
}
private String hashPassword(String password) { 5 usages
Money > app > src > main > java > com > example > money > managers > UserPreferencesManager > register
50:1 CRLF UTF-8 4 spaces


UserPreferencesManager.java
public class UserPreferencesManager { 12 usages
public boolean setNewPassword(String newPassword) { 1 usage
SharedPreferences.Editor editor = sharedPreferences.edit();
editor.putString(KEY_PASSWORD, hashedPassword);
editor.putBoolean(KEY_HAS_PASSWORD, true);
editor.putBoolean(KEY_IS_LOGGED_IN, true);
return editor.commit();
}
private String hashPassword(String password) { 5 usages
try {
MessageDigest digest = MessageDigest.getInstance(algorithm: "SHA-256");
byte[] hash = digest.digest(password.getBytes(StandardCharsets.UTF_8));
return Base64.encodeToString(hash, Base64.NO_WRAP);
} catch (NoSuchAlgorithmException e) {
e.printStackTrace();
return null;
}
}
}
Money > app > src > main > java > com > example > money > managers > UserPreferencesManager > register
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